

Dear Editors,

Please find enclosed our submission entitled

“Boyd’s conductor-11 Mahler measure conjecture: proof of the split-integral identity (C3), with an exact structural analysis of the family S_k ”

by Huimin Zheng (College of Information and Network Engineering, Anhui Science and Technology University, Fengyang, Anhui 233100, P. R. China; zhhm@ahstu.edu.cn), which we submit for consideration in *Experimental Mathematics*.

This manuscript is original, has not been published previously, and is not under consideration by any other journal. All authors (a single author) are aware of and approve this submission.

Summary and significance. This is a story of experimental mathematics carried through to a rigorous proof. Boyd’s 1998 numerical experiments produced conjectural identities between Mahler measures of two-variable polynomials and L -values of elliptic curves, beginning with the smallest possible conductor, $N = 11$. Of the three conductor-11 identities, the first two were proved by Brunault; the third, (C3), concerns a polynomial vanishing on the unit torus and asserts that a signed split integral of $\log |y|$ around the branch cut equals $b_{11} = L'(E_{11}, 0)$. This paper proves (C3), completing the conductor-11 block of Boyd’s table. Highlights:

- **Computation-driven proof:** the proof strategy — closing Samart’s signed open chain by a small-branch compensating arc — was found through numerical experimentation; the crucial homology statement (period ratio equal to exactly 2) is then pinned down by certified interval arithmetic (Arb ball arithmetic with rigorous error bounds), not merely observed numerically.
- **Full reproducibility:** every certification step is carried out within interval arithmetic, and all scripts, frozen certificates, and run logs are openly available at <https://github.com/huiminZheng-collab/boyd-conductor11> and archived at <https://doi.org/10.5281/zenodo.21820650>.
- **New experimental results:** a complete structural analysis of the family S_k , new Boyd-type numerical identities (recorded as conjectures, to up to 70 digits, PARI/GP cross-checked), and a mechanistic analysis of the conductor-53 case showing precisely why the closed-cycle/modular-unit mechanism provably fails there — a negative result that delimits the method.
- A direct regulator computation via Brunault’s proved Siegel-unit formula bypasses Bloch’s diamond theorem and its normalization ambiguities.

We believe this combination of conjecture discovery, certified computation, structural classification, and mechanism-level counterexample analysis is exactly the kind of work *Experimental Mathematics* aims to publish.

Transparency. The research was carried out by the author with the assistance of the AI system Kimi (Moonshot AI); a formal Declaration on the use of AI tools appears on the title page of the article, and the author has verified all mathematical content and takes full responsibility for it. A preprint has been submitted to arXiv (identifier to be announced; available on request).

Suggested referees.

- Matilde N. Lalin (Université de Montréal)
- François Brunault (ENS de Lyon)

- Mathew Rogers
- Wadim Zudilin (Radboud University)
- Detchat Samart
- Marie-José Bertin

The author has no conflicts of interest with the suggested referees.

Thank you for your consideration.

Sincerely,

Huimin Zheng
Anhui Science and Technology University
zhhm@ahstu.edu.cn